

Hypothesis testing with multiple regression

Last time: Class activity

```
anova(m1)
```

Response: Height

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
Age	1	5005.2	5005.2	575.820	< 2.2e-16	***
Sex	1	295.8	295.8	34.035	2.235e-08	***
Age:Sex	1	227.2	227.2	26.136	7.599e-07	***
Residuals	194	1686.3	8.7			

```
anova(m2)
```

Response: Height

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
Sex	1	480.7	480.67	13.99	0.0002412	***
Residuals	196	6733.9	34.36			

$$F = \frac{\frac{1}{\# \text{ parameters tested}} (SSE_{reduced} - SSE_{full})}{\frac{1}{n-p} SSE_{full}}$$

Class activity

What is the p-value for this test?

```
pf(290.35, df1 = 2, df2 = 194, lower.tail=F)
```

```
## [1] 4.686114e-59
```

The p-value is basically 0, so we have strong evidence that there is a relationship between Age and Height, after accounting for Sex.

Testing in R

We can do comparisons between nested models a little more efficiently:

```
m1 <- lm(Height ~ Age*Sex, data = Kids198)
m2 <- lm(Height ~ Sex, data = Kids198)

anova(m2, m1)
```

```
## Analysis of Variance Table
```

```
##
```

```
## Model 1: Height ~ Sex
```

```
## Model 2: Height ~ Age * Sex
```

```
##      Res.Df      RSS Df Sum of Sq      F      Pr(>F)
```

```
## 1         196 6733.9
```

```
## 2         194 1686.3  2      5047.6 290.35 < 2.2e-16 ***
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1
```

Testing a single parameter

$$\text{Height} = \beta_0 + \beta_1 \text{Age} + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

$$H_0 : \beta_3 = 0 \quad H_A : \beta_3 \neq 0$$

How did we test a single coefficient in simple linear regression?

Testing a single parameter

$$\text{Height} = \beta_0 + \beta_1 \text{Age} + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

$$H_0 : \beta_3 = 0 \quad H_A : \beta_3 \neq 0$$

How did we test a single coefficient in simple linear regression?

With a t-test!

Testing a single parameter

$$\text{Height} = \beta_0 + \beta_1 \text{Age} + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

$$H_0 : \beta_3 = 0 \quad H_A : \beta_3 \neq 0$$

$$t = \frac{\hat{\beta}_3 - 0}{SE_{\hat{\beta}_3}}$$

Under H_0 , this test statistic follows a t distribution.

What do you think should be the degrees of freedom of the t distribution?

Testing a single parameter

$$\text{Height} = \beta_0 + \beta_1 \text{Age} + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

$$H_0 : \beta_3 = 0 \quad H_A : \beta_3 \neq 0$$

$$t = \frac{\hat{\beta}_3 - 0}{SE_{\hat{\beta}_3}}$$

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What do you think should be the degrees of freedom of the t distribution?

$$n - p$$

Testing a single parameter

$$\text{Height} = \beta_0 + \beta_1 \text{Age} + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

$$H_0 : \beta_3 = 0 \quad H_A : \beta_3 \neq 0$$

$$t = \frac{\hat{\beta}_3 - 0}{SE_{\hat{\beta}_3}}$$

Under H_0 , this test statistic follows a t_{n-p} distribution.

Testing a single parameter

$$\text{Height} = \beta_0 + \beta_1 \text{Age} + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

```
m1 <- lm(Height ~ Age*Sex, data = Kids198)
```

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	32.960916	1.537781	21.434	< 2e-16	***
Age	0.182543	0.009383	19.454	< 2e-16	***
Sex	-7.727846	2.034957	-3.798	0.000195	***
Age:Sex	0.064134	0.012545	5.112	7.6e-07	***

$$t = \frac{\hat{\beta}_3 - 0}{SE_{\hat{\beta}_3}} = \frac{0.0641}{0.0125} = 5.112$$

$$\text{p-value} = 7.6 \times 10^{-7}$$

Testing a single parameter

$$\text{Height} = \beta_0 + \beta_1 \text{Age} + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

Research question: Is the slope on *Age* *larger* for boys than for girls?

What hypotheses could I test to answer this research question?

Testing a single parameter

$$\text{Height} = \beta_0 + \beta_1 \text{Age} + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

Research question: Is the slope on *Age* *larger* for boys than for girls?

What hypotheses could I test to answer this research question?

$$H_0 : \beta_3 = 0 \quad H_A : \beta_3 > 0$$

Test statistic: $t = \frac{\hat{\beta}_3 - 0}{SE_{\hat{\beta}_3}} = 5.112$

Which tail do I use to calculate the p-value?

Testing a single parameter

$$\text{Height} = \beta_0 + \beta_1 \text{Age} + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

Research question: Is the slope on Age *larger* for boys than for girls?

$$H_0 : \beta_3 = 0 \quad H_A : \beta_3 > 0$$

Test statistic: $t = \frac{\hat{\beta}_3 - 0}{SE_{\hat{\beta}_3}} = 5.112$

```
pt(5.112, df = 194, lower.tail=F)
```

```
## [1] 3.805925e-07
```

p-value = 3.8×10^{-7} , so we have strong evidence that the slope on Age is larger for boys than for girls.

Testing a single parameter

$$\text{Height} = \beta_0 + \beta_1 \text{Age} + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

Research question: Is the slope on *Age* *larger* for boys than for girls?

$$H_0 : \beta_3 = 0 \quad H_A : \beta_3 > 0$$

Can I use a nested F test here?

No -- a nested F test doesn't let me test a specific direction ($\beta_3 > 0$)

t tests vs. nested F tests

t test: test a single parameter

Some example hypotheses:

+

$$H_0 : \beta_3 = 0 \quad H_A : \beta_3 \neq 0$$

+

$$H_0 : \beta_3 = 0 \quad H_A : \beta_3 > 0$$

+

$$H_0 : \beta_3 = 1 \quad H_A : \beta_3 < 1$$

nested F test: test one or more parameters

Some example hypotheses:

$$H_0 : \beta_1 = \beta_2 = 0$$

$$H_A : \text{at least one of } \beta_1, \beta_2 \neq 0$$

Confidence intervals

Just like with simple linear regression, we can construct confidence intervals for single parameters.

$$\text{Height} = \beta_0 + \beta_1 \text{Age} + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

Confidence interval for β_3 :

Confidence intervals

Just like with simple linear regression, we can construct confidence intervals for single parameters.

$$\text{Height} = \beta_0 + \beta_1 \text{Age} + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

Confidence interval for β_3 :

$$\hat{\beta}_3 \pm t^* SE_{\hat{\beta}_3}$$

where t^* is the critical value for a t_{n-p} distribution.

Confidence intervals

$$\text{Height} = \beta_0 + \beta_1 \text{Age} + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

Confidence interval: $\hat{\beta}_3 \pm t^* SE_{\hat{\beta}_3}$

Suppose I want to calculate a 99% confidence interval. How do I find the critical value t^* ?

Confidence intervals

$$\text{Height} = \beta_0 + \beta_1 \text{Age} + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

Confidence interval: $\hat{\beta}_3 \pm t^* SE_{\hat{\beta}_3}$

Suppose I want to calculate a 99% confidence interval. How do I find the critical value t^* ?

```
qt(0.005, df=194, lower.tail=F)
```

```
## [1] 2.601409
```

$$t^* = 2.60$$

Confidence intervals

$$\text{Height} = \beta_0 + \beta_1 \text{Age} + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

Confidence interval: $\hat{\beta}_3 \pm t^* SE_{\hat{\beta}_3}$

$$0.0641 \pm 2.60 \cdot 0.0125 = (0.0316, 0.0966)$$

So we are 99% confident that a one-month increase in age is associated with an change in height that is between 0.0316 and 0.0966 inches greater for boys than for girls.